

## Technical Skills

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- Programming Languages: **JavaScript**, TypeScript, Python, C/C++, **C#**, Java, Golang,
- Web Technologies: **React.js**, HTML5, CSS, Node.js, Ant Design, moment.js
- Game Development: **Unity**, **Unreal Engine**, SDL
- Database & Tools: **MongoDB**, **SQL**, Git, VS Code, Visual Studio
- Security & Infra Tools: Scapy, GDB, Wireshark, Vagrant, Metasploit, Terraform, **AWS CLI**

## Technical Experience

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**JavaScript Developer Co-op** **Semios** **Jan 2022 – Aug 2022**

- Built and maintained **React** components following **Agile** methodologies, ensuring consistent delivery quality.
- Built reusable UI components using **Ant Design** library, implementing features like **data tables**, forms, and **interactive map interfaces**.
- Resolved front-end bugs and implemented UI/UX improvements through effective collaboration with **QA team**.
- Enhanced application functionality by implementing robust **form validation**, content sorting, and date/time management using **moment.js**.

**Web Developer** **Bluvec Technologies Inc.** **Sep 2021 – Dec 2021**

- Developed company's landing page using **React**, ensuring responsive design and brand consistency.
- Implemented automated testing suite using **Golang** to improve code reliability.
- Built automated image management system using **GoCV** for timestamp processing and file cleanup.

## Technical Projects

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**APEX** **Python, MCP** **Jan 2026 - Present**

- Built an autonomous AI-driven penetration testing framework evaluated on **42 HackTheBox machines**, achieving a **71% success rate** (30/42) across Linux and Windows targets.
- Built a multi-agent pipeline where recon (nmap, gobuster, ZAP) and execution agents dispatch LLM-generated bash scripts to Kali Linux via a custom MCP-to-HTTP REST API bridge.
- Designed an LLM-based **failure classifier** (FIXABLE / FUNDAMENTAL / SCRIPT\_ERROR) with three independent loop-detection counters to prevent infinite retry cycles and auto-fail stalled tasks.
- Implemented dual **RAG pipelines** — offensive payload retrieval and historical failure retrieval — to guide exploit selection and avoid previously failed attack paths.

**Music Stats Page** **Next.js, React, Tailwind CSS, Redis** **June 2025**

- Integrated APIs from Last.fm, iTunes, and Spotify to fetch top tracks/artists and enrich album artwork and links
- Built **responsive** React components display listening history with fallback handling
- Implemented **Redis caching** with 1-hour TTL to reduce API calls and support manual/automatic cache refresh
- Optimized data fetching using Promise.all and normalized JSON structures from multiple APIs

**TCP Network Attack Simulation** **Python, Scapy, Vagrant, GDB** **April 2025**

- Simulated and evaluated **TCP-layer attacks** in a 3-node virtual network using Vagrant.
- Launched a **TCP SYN flood** to exhaust server backlog and observed mitigation via SYN cookies.
- Executed a **TCP RST injection** to forcibly terminate active Telnet/SSH sessions.
- Crafted spoofed packets to **hijack Telnet sessions**, inject shell commands, and validate impact via packet capture and file system changes.

**ROP Exploitation** **C, Python, GDB, ROPgadget** **Mar 2025**

- Developed and tested **ROP chains** against a custom x64 Linux binary with **NX protection enabled and ASLR disabled**.
- Used **GDB and ROPgadget** to locate gadgets and construct precise exploit payloads, including handling register constraints and gadget chaining.
- Wrote Python scripts to automate ROP payload generation and gadget chaining using offsets from libc.

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| <b>PrediQL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Python, GraphQL, Ollama</b>                    | <b>Sep 2024 – Dec 2024</b> |
| <ul style="list-style-type: none"> <li>Developed an AI-assisted fuzzing pipeline targeting <b>GraphQL endpoint vulnerabilities</b> such as <b>SQL Injection, DoS, and batching attacks</b>.</li> <li>Designed Python scripts to automate <b>LLaMA-based query generation</b> via <b>Ollama</b>.</li> <li>Converted LLaMA outputs into structured JSON payloads and used them to test GraphQL APIs, iteratively refining queries based on error responses.</li> </ul>                                                                                                                                                                          |                                                   |                            |
| <b>Cloud Infrastructure Deployment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>EC2, VPC, IAM, DynamoDB</b>                    | <b>Nov 2024</b>            |
| <ul style="list-style-type: none"> <li>Deployed a secure VPC network on AWS using the web console, including <b>public/private subnets, Internet Gateway, NAT Gateway, Route Tables, and Security Groups</b>.</li> <li>Launched <b>EC2 instances</b> and configured a <b>Node.js web server</b> accessible on port 8081.</li> <li>Created and managed <b>IAM users, groups, and roles</b>, applying the principle of least privilege.</li> <li>Built a <b>DynamoDB</b> table and securely connected it to EC2 via role-based permissions.</li> <li>Implemented <b>Elastic IPs</b> and studied their behavior on instance restarts.</li> </ul> |                                                   |                            |
| <b>Serverless API Development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>AWS Lambda</b>                                 | <b>Nov 2024</b>            |
| <ul style="list-style-type: none"> <li>Built a <b>serverless REST API</b> using <b>AWS Lambda</b> and <b>API Gateway</b>, eliminating the need for EC2 servers.</li> <li>Developed a <b>Lambda function</b> to support both <b>GET and POST</b> requests for a comment system.</li> <li>Secured the <b>POST</b> endpoint via <b>AWS Cognito</b>, implementing <b>JWT-based authorization</b>.</li> <li>Validated API with Postman, tested user-based access control, and discussed <b>JWT tamper-resistance mechanisms</b>.</li> </ul>                                                                                                        |                                                   |                            |
| <b>Eggly Buddy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>C++, OpenGL, ECS Framework</b>                 | <b>Sep 2023 – Dec 2023</b> |
| <ul style="list-style-type: none"> <li>Programmed interactive elements using <b>C++</b>, enabling in-game pet mechanics and mini-games.</li> <li>Implemented visual effects using custom shaders, including dynamic parallax background and animations.</li> <li>Developed save/load system using <b>nlohmann/json</b> library for persistent game state.</li> </ul>                                                                                                                                                                                                                                                                          |                                                   |                            |
| <b>Swag the North</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Unity, C#</b>                                  | <b>Sep 2023</b>            |
| <ul style="list-style-type: none"> <li><b>Top 12 Finalist</b> out of 200+ projects in Hack the North.</li> <li>Spearheaded development of an immersive game using <b>Unity</b> and <b>C#</b>, featuring innovative swag collection mechanics.</li> <li>Implemented core gameplay systems including player controls, item spawning, and scene management.</li> <li>Led version control workflow and mentored team members in Git best practices.</li> </ul>                                                                                                                                                                                    |                                                   |                            |
| <b>Locative Audio</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Unity, C#</b>                                  | <b>Jan 2023 – Apr 2023</b> |
| <ul style="list-style-type: none"> <li>Developed a <b>GPS-based</b> mobile app that lets users discover Vancouver’s cultural audio content at specific locations.</li> <li>Built an efficient content management system using <b>XML</b> data structure and integrated <b>AR Foundation</b> for image detection.</li> <li>Created an interactive map interface with <b>iGUI</b> featuring custom camera controls and smooth user navigation.</li> </ul>                                                                                                                                                                                       |                                                   |                            |
| <b>FREN</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>React, JavaScript, MongoDB, Express, Redux</b> | <b>Jun 2021 – Aug 2021</b> |
| <ul style="list-style-type: none"> <li>Built a social networking app with <b>user recommendation system</b> and <b>real-time chat</b> functionality.</li> <li>Implemented <b>Redux</b> state management and routing for seamless user experience.</li> <li>Developed responsive chat interface with <b>custom pop-up components</b> for friend recommendations.</li> </ul>                                                                                                                                                                                                                                                                    |                                                   |                            |
| <b>Education</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |                            |
| <b>Vancouver, BC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Simon Fraser University</b>                    | <b>Sep 2024 – May 2026</b> |
| <ul style="list-style-type: none"> <li>Master’s in Professional Computer Science (Focus: <b>Cybersecurity</b>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                   |                            |
| <b>Vancouver, BC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>University of British Columbia</b>             | <b>Sep 2018 – May 2024</b> |
| <ul style="list-style-type: none"> <li>Graduated with a BS in Computer Science, <b>with Distinction</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |                            |